

HENRY HSU

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SUMMARY

Former GenAI Team Lead with experience building production GenAI platforms that doubled the B2B client base and grew the user base to 20K. Built key product features including Agentic AI Systems, Advanced RAG, and Backend infrastructure, leading to two major industry awards. Now pursuing a Master of Computer Science at Rice University.

WORK EXPERIENCE

Generative AI Team Lead

Dec. 2024 – Aug. 2025

MaiAgent Co., Ltd. – Award-Winning B2B GenAI Startup

Taipei, Taiwan

- **Building AI Agents** – Upgraded chatbots into AI agents by integrating memory systems, tool APIs, and MCP Client, resulting in **120%** partner growth (CTBC Bank, MSI, HPE, iGroup) and user expansion from **3K** to **20K** users while cutting LLM token usage by **67%+**
- **Contributing to Open Source** – Contributed **14** merged pull requests to the **LlamaIndex** open-source project (**15K+ stars**), fixing bugs and adding features in integrations with AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, MCP Client, and AI Agent Workflow
- **Testing Framework** – Introduced pytest testing framework for unit and E2E coverage to prevent production regressions, achieving **67%** test coverage from **zero** baseline, which reduced production hotfixes by **90%** initially and sustaining a **50%** reduction long term
- **Technical Roadmap & Leadership** – Defined GenAI product roadmap for a 10-person startup; led feasibility evaluation and end-to-end delivery of production features including Agentic RAG, Artifact Generation, Tagging & Permission System, and Information Retrieval

Generative AI Intern

Sep. 2024 – Dec. 2024

- **Pipeline Scaling** – Refactored with async framework and dual-database synchronization (RDB/Vector DB), achieving **3.5x** faster parsing and scaling from **3M** to **20M+** text chunks while **doubling** indexing performance
- **Reranker Integration** – Integrated Cohere and BGE open-source rerankers as a customer-selectable retrieval feature, improving RAG precision on large documents by **30% Precision@5** and **7% Precision@10**
- **Citation Architecture** – Redesigned parsing pipeline by decoupling Document and TextNode layers, enabling precise citations

EDUCATION

Rice University - Top 20 U.S. University

Aug. 2025 – Dec. 2026 (Expected)

M.C.S., Computer Science (GPA: 4.00/4.00)

Houston, TX

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University

Sep. 2020 – Jun. 2024

Minor: Computer Science (Domain GPA: 4.13/4.30)

Hsinchu, Taiwan

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List)

PROJECTS

Agentic Hybrid RAG | Graph RAG (Neo4j, Cypher), Vector RAG (Milvus), Multi-Agent (LangGraph)

Feb. 2024 – Aug. 2024

- **Self-Routing Agentic RAG**: Implemented a query classification agentic AI system that picks a global or local approach to request slicing from Graph RAG, Vector RAG or a Hybrid (Graph & Vector) RAG path to answer the contextual or detail-oriented queries
- **ML Clustering**: Applied ML clustering algorithms (K-means and DBSCAN) in recognizing pivotal nodes from vector database to build a low-cost Graph RAG; saved **35%** of indexing token cost, with only 2.56% worse in RAGAS metrics
- **Advanced Hybrid RAG**: Enhanced vector RAG hit rate using data retrieved from Graph RAG with HyDE, balancing both global and detailed information for more accurate responses. Achieved an MAR@10 of **88.2%** on multi-hop datasets

GenAI Tutoring System with Multi-Agent RAG | LangChain, CrewAI, Streamlit, Chroma

Feb. 2024 – Jul. 2024

- **Competition Achievement**: **Top 3** of 50 teams in the Computer Science AI workshop, winning the Outstanding Award
- **Multi-Agent Framework**: Utilized **CrewAI** to assign roles to LLMs for collaborative problem-solving, assisting students with problem-solving, exam detection, concept analysis, and compared open-source LLMs (Llama 3) and commercial LLMs (GPT-4)
- **Interactive UI**: Enabled user tuning of LLM settings and inspection of chain-of-thought from ReAct Agent and citations by **Streamlit**

AWARDS

Presidential Hackathon Winner (2024) - Urban noise detection with LLM-powered structured analysis **23rd Golden Peak Award (2025)**

- Outstanding Commercial Product, MaiAgent AI Platform

SKILLS

Languages: Python, C/C++, TypeScript | **GenAI**: LlamaIndex, LangChain, LangGraph, MCP | **Fullstack**: Django, FastAPI, Express.js, React

Databases: PostgreSQL, MongoDB, Milvus, Elasticsearch | **Tools**: nvim, git, Docker, Shell Script, Nginx | **Methodologies**: Agile, Scrum

Data Science: numpy, pandas, scikit-learn, PyTorch, JAX | **AWS**: Bedrock, EMR, EC2, S3, RDS, ElastiCache, SageMaker